

School of Computer Science
COMP 2067 Programming for Beginners
Summer 2025 – Lab 4
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Logical and Relational Operators and Decision Structures

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OBJECTIVES:

- i. To understand logical and relational operators.
- ii. To test conditions to see if they are True or False.
- iii. To use simple decision structures (if-else) in a program.

1. Launch **IDLE** in your computer, open a **new file** through IDLE, and name it *lab4sol.py*.

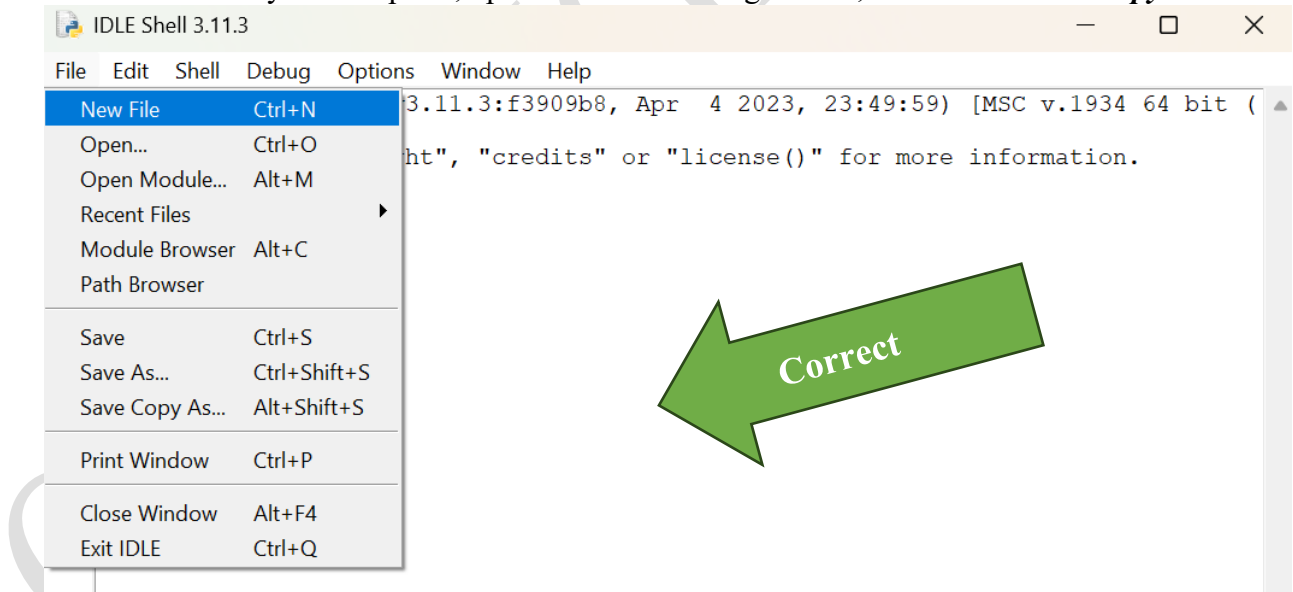


Fig. 1: Creating a new file

2. In *lab4sol.py*, write a Python program **with comments** to do the following:
 - a. Ask the user to enter a password containing at least 8 characters. Assign the entered string to a variable *password*.
 - b. Check the length of *password*:
 - i. If *password* has less than 8 characters, print: "<password> is too short to be a valid password."

- ii. If password consists only of alphabetic characters, print: “Your password is weak because it contains only letters.”
 - c. Otherwise (if password contains numbers or special characters), do the following:
 - i. Print a statement: “*<password>* is a valid complex password.”
 - ii. Find the index of the first occurrence of the digit ‘5’ in *password* and print this value with a *suitable message*¹. A value of -1 indicates *password* does not contain ‘5’.
 - d. Find the total number of characters in *password*² and print it with a *suitable message*¹.
3. **Submit your Python code** (i.e., as lab4sol.py file) through **Brightspace**.

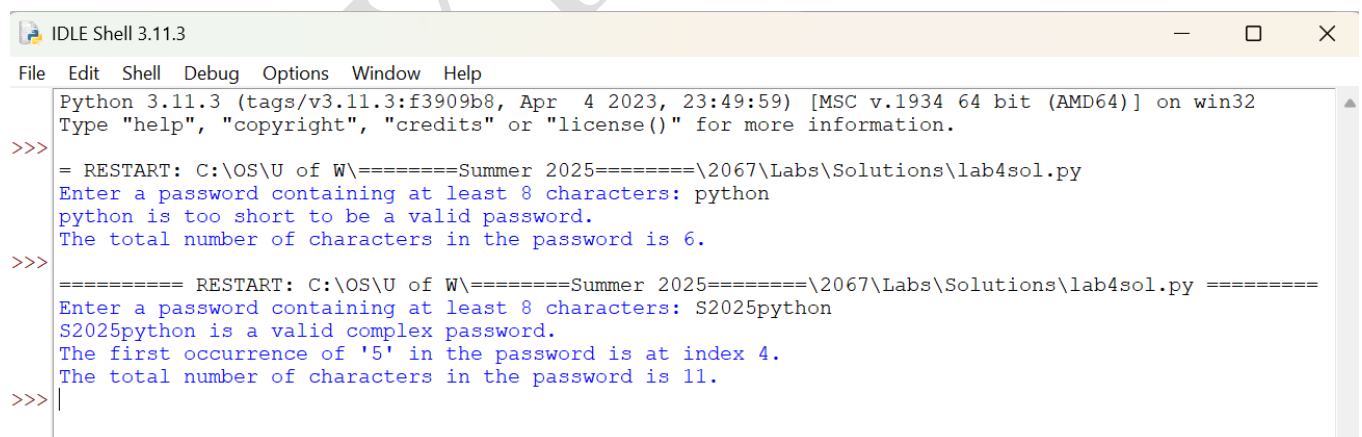
¹**NOTE:** In this course, the phrase ‘suitable message’ refers to some kind of description that describes what is being printed. For example, just printing the value **22** by itself does not tell the user anything useful; however, printing ‘**x = 22**’ would add some context to what is being printed and so this would qualify as a ‘suitable message’ if the value of x is to be printed.

²**NOTE:** **step 2d** should be performed for all cases, i.e. it should be done whether *password* is a valid password (i.e. has at least 8 characters) or not.

³**NOTE:** **Do NOT submit a screenshot or the contents of the IDLE shell displaying your output. These will NOT be marked.** You must submit the *lab4sol.py* file.

Below is a **sample** screenshot showing example outputs for Lab 4.

Please note: do not submit a screenshot; only submit your lab4sol.py file. Your suitable message does not need to match the screenshot exactly.



```

Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\OS\U of W\=====Summer 2025=====\\2067\Labs\Solutions\lab4sol.py
Enter a password containing at least 8 characters: python
python is too short to be a valid password.
The total number of characters in the password is 6.
>>>
===== RESTART: C:\OS\U of W\=====Summer 2025=====\\2067\Labs\Solutions\lab4sol.py =====
Enter a password containing at least 8 characters: S2025python
S2025python is a valid complex password.
The first occurrence of '5' in the password is at index 4.
The total number of characters in the password is 11.
>>>
  
```

Fig. 2. A sample screenshot showing example outputs for Lab 4.

Submission:

1. You will earn a maximum of 100 points (accounts for 2.5%) for submitting your **source code** (*lab4sol.py*) within the specified deadline.
2. **Do not upload your source code to Brightspace as a single zip/rar file.** Such submissions will not be accepted and a mark of **zero** will be assigned.

3. Marks will be **deducted** if a *suitable message and/or comments* are missing.
4. Late submissions will incur a penalty of 10% per day for **up to two days**. Submissions made beyond two days will not be accepted, and a mark of **zero** will be assigned.
5. Unlimited resubmissions are allowed. But keep in mind that we will consider the last submission. That means that if you resubmit after the deadline, a penalty will be applied, even if you submitted an earlier version on time.

Academic Integrity.

Plagiarism is a serious offense and can result in severe consequences such as receiving a grade of **zero** on the assignment/lab or even being asked to leave the program.

Copying or using someone else's code is considered plagiarism. This includes using code from online sources, previous labs/assignments, or from other students. Even if you have modified the code, our antiplagiarism software can still show it as plagiarism.

To avoid plagiarism, make sure to always use your own words and ideas when writing source code. Additionally, always check with your instructor to make sure you understand what is allowed and what is not in terms of using outside resources.

Remember that academic integrity is essential for your own personal and professional growth, and it is your responsibility to uphold these principles. So please take it seriously and always produce your own original work.